

Module/Pathway Review: Choline-O-sulfate uptake and desulfation in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Pathway/module:** choline-O-sulfate-utilization (free-text; no local module YAML or candidate gene table supplied) **Curation intent:** module satisfiability + gene-annotation review **Evidence date:** 2026-08-31

1. Executive summary

P. putida KT2440 is genomically equipped to import and desulfate choline-O-sulfate (COS), and the module should be marked **satisfiable (largely covered)**. The pathway core is a discrete, isolated three-gene cassette on the chromosome:

- **PP_0077 = *betC*, choline-sulfatase (EC 3.1.6.6)** — the desulfation step (COS → choline + inorganic sulfate). **COVERED** (high-confidence homology + catalytic-motif evidence).
- **PP_0076** — OpuAC/ProX-family periplasmic **choline/COS-binding protein** (uptake). **COVERED, but transporter architecture is ambiguous.**
- **PP_0075** — annotated "choline sulfate transporter" but is actually a **SulP/SLC26 anion (sulfate) permease**; the specific "choline sulfate" label is a **likely over-annotation**. Uptake mechanism = **candidate_uncertain**.

A candidate pathway-specific regulator (**PP_0079**) sits immediately downstream. BetC is a **formylglycine-dependent type-I sulfatase** (canonical C-x-P-x-R motif), so it depends on the maturation enzyme **PP_3353 (formylglycine-generating enzyme, FGE)** — an obligatory accessory step. The liberated **choline** feeds the well-annotated betaine pathway (*betA/betB*), and the liberated **sulfate** enters standard cysteine/sulfur assimilation.

Important caveat: essentially all target-taxon evidence is **computational/homology-based**. There is, to our knowledge, **no direct experimental characterization of COS utilization in KT2440**. Functional transfer comes from *Sinorhizobium meliloti* (BetC; strong, same enzyme family) and *Bacillus subtilis* (OpuC-mediated COS uptake; mechanistically informative but transporter-family-divergent).

2. Target-organism pathway definition

Included process (pathway boundary): the *uptake of the plant/fungal-derived osmolyte choline-O-sulfate from the periplasm/environment, followed by intracellular hydrolysis of its sulfate-ester bond to yield free choline + inorganic sulfate*. The module's two committed steps are (i) **COS import** and (ii) **COS desulfation (BetC)**. Sulfate then serves as a **sulfur source**; choline is a **carbon/nitrogen source and osmoprotectant precursor**.

Neighboring pathways to keep separate (do NOT merge into this module): - **Choline → glycine betaine oxidation** (*betA* PP_5064, *betA-I* PP_0056 choline dehydrogenase EC 1.1.99.1; *betB* PP_5063 betaine aldehyde dehydrogenase EC 1.2.1.8) — a *downstream* module that consumes the choline product; it is separately annotated and much better supported. - **Glycine-betaine catabolism** (*gbcAB* PP_0315/PP_0316 dioxygenase → dimethylglycine ...) — further downstream C/N mobilization. - **Osmoprotectant/quaternary-amine transport** (BCCT carriers *betT-I/II/III* PP_5061/PP_0229/PP_3957; ABC systems *cbcVWX* PP_0294–0296, *opuA* PP_2774, *yehWXYZ* PP_0868–0871) — broad choline/betaine/carnitine transport overview; overlaps but is not COS-specific. - **Assimilatory sulfate reduction / sulfonate & sulfate-ester scavenging** (Cys pathway; *atsA* PP_3352 arylsulfatase; alkylsulfatase PP_2045) — receives the sulfate product but is a different pathway. Keep the general "sulfur-limitation ester scavenging" map separate from the choline-specific COS module.

Alternate names / database definitions: choline-O-sulphate; choline sulfate; EC 3.1.6.6 = *choline-sulfatase*; MetaCyc/BioCyc frames COS desulfation within "choline degradation"/"glycine betaine biosynthesis" supersets; KEGG lists *ppu:PP_0077* with the K-ortholog for choline-sulfatase. InterPro family **IPR017785 (Choline-sulfatase)** is the tightest definition and cleanly distinguishes BetC from generic arylsulfatases.

3. Expected step model

Step	Expected function	Best KT2440 candidate	Call
S1	COS import across inner membrane	PP_0075 (SulP permease) and/or PP_0076 (ABC SBP) + a QA-ABC permease/ATPase	candidate_uncertain
S2	COS desulfation → choline + sulfate	PP_0077 (betC, EC 3.1.6.6)	covered
S2a	Sulfatase maturation (Cys→formylglycine)	PP_3353 (FGE)	covered (accessory, obligatory)
S3	Fate of choline (oxidation to betaine)	<i>betA</i> PP_5064 / PP_0056; <i>betB</i> PP_5063	covered (separate module)
S4	Fate of sulfate (assimilation)	Cys/sulfate-assimilation genes	covered (separate module)
R	Pathway-specific transcriptional regulation	PP_0079 (adjacent regulator)	candidate

4. Candidate genes and evidence

PP_0077 — *betC*, Choline-sulfatase (Q88RQ2; EC 3.1.6.6; 505 aa). *High confidence.* - Evidence: InterPro **IPR017785 (dedicated Choline-sulfatase family)** + IPR000917 Sulfatase_N + IPR025863 (choline-sulfatase C-domain), Pfam PF00884 + PF12411; GO:0047753 choline-sulfatase activity. UniProt proteinExistence = "3: Inferred from homology." - Sequence-level confirmation: canonical **type-I sulfatase catalytic motif C51-A-P-S-R55** (...AYCAPSRFTL...), the Cys that is oxidized to Cα-formylglycine. - Curation caveat: no direct KT2440 assay; functional transfer from *S. meliloti* BetC

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is **strong** (same InterPro family, same motif, syntenic role). **Promote to full review.** [fetch-gene](#)

PP_0076 — Choline/betaine-binding protein (Q88RQ3; 307 aa). *Medium confidence for uptake, low for COS specificity.* - Evidence: Pfam **PF04069 OpuAC**, InterPro IPR017783 (ABC choline substrate-binding) / IPR007210 (ABC glycine-betaine SBP); GO:0033265 choline binding, GO:0015871 choline transport; signal peptide; proteinExistence "4: Predicted." - Caveat: it is the **periplasmic SBP of an ABC importer, but the cognate permease/ATPase are not in the immediate cluster** (orphan SBP). It may pair with a shared QA-ABC permease/ATPase (e.g., *yehWXYZ* PP_0868–0871 or *cbcVW* PP_0294–0295). Whether it binds COS specifically vs. choline is unproven. Paralog ambiguity with *cbcX* PP_0296, *betX* PP_1741, PP_0870.

PP_0075 — "Choline sulfate transporter" (Q88RQ4; 521 aa). *Likely mis-specific annotation.* - Evidence: InterPro **IPR001902/IPR011547 (SLC26A/SulP family)** + STAS domain (PF01740) + PF00916 Sulfate_transp; GO:0055085 transmembrane transport; proteinExistence "4: Predicted." This is a **SulP/SLC26 anion (typically inorganic sulfate) permease**, not a BCCT/QA carrier. - Interpretation: the specific "choline sulfate" substrate name is **probable over-annotation derived from gene neighborhood** (it abuts *betC*). Plausible real roles: import of the anionic COS ester, or import/recycling of the inorganic **sulfate product**. **Substrate = uncertain; promote to full review and correct the substrate-specific label.**

PP_0079 — Transcriptional regulator (Q88RQ0; 299 aa). Adjacent to the cassette; candidate local regulator (analogous to BetI in the *S. meliloti bet* operon). Not yet assigned a family here — worth a `fetch-gene` review.

PP_3353 — Sulfatase-modifying factor / FGE (Q88HK3; 341 aa). Obligatory maturation enzyme for BetC (and for *atsA* PP_3352, alkylsulfatase PP_2045). Should be recorded as an **accessory module dependency**, not a per-organism gap.

Downstream (context, separate modules): *betA* PP_5064 & *betA-I* PP_0056 (choline dehydrogenase), *betB* PP_5063 (betaine aldehyde dehydrogenase) — high-confidence, HAMAP-ruled; consume the choline product.

5. Gaps, ambiguities, and likely over-annotations

- **Likely over-annotation (flag):** PP_0075 "choline sulfate transporter" — family is SulP/SLC26; substrate specificity for choline-O-sulfate is unsupported. Recommend relabeling to "SulP-family sulfate/anion permease (putative COS/sulfate transporter)."
- **Ambiguity — which transporter actually imports COS:** three non-exclusive possibilities: (a) PP_0075 SulP permease imports the COS anion; (b) PP_0076 ABC-SBP (with a shared QA-

ABC permease/ATPase) imports COS as a choline ester; (c) broad BCCT/OpuC-like systems co-import COS (as in *B. subtilis*, where OpuC handles COS;

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Step S1 = candidate_uncertain.

- **PP_0075 is probably not redundant inorganic-sulfate transport:** KT2440 already encodes a dedicated assimilatory sulfate ABC importer (*cysA* PP_5168 [EC 7.3.2.3], *cysW* PP_5169, sulfate-binding *sbp-I* PP_4305 / *sbp-II* PP_5171). Because inorganic-sulfate uptake is already covered by that system, the SulP permease PP_0075 embedded in the COS cassette is more plausibly dedicated to importing the anionic **COS ester** (or recycling the liberated sulfate) — modestly raising confidence that PP_0075 belongs to the uptake step, though its substrate remains experimentally unproven.
- **PP_0078** has no distinct UniProt entry (small/hypothetical ORF); the cassette effectively reads PP_0075–PP_0076–PP_0077 followed by regulator PP_0079.
- **Orphan SBP:** PP_0076 lacks an adjacent permease/ATPase — a genuine module-structure question.
- **No direct KT2440 phenotype:** growth on COS as sole S (or C/N) source, and *betC*/PP_0075/PP_0076 mutant phenotypes, have not been reported for KT2440. All calls are homology/motif-based.
- **Not a gap:** the desulfation step itself is well covered; do **not** mark the module as a gap.

6. Module and GO-curation recommendations

- **S2 desulfation → COVERED** by PP_0077 (*betC*, EC 3.1.6.6). GO:0047753 (choline-sulfatase activity) is appropriate and well-supported.
- **S2a maturation → COVERED (accessory):** add PP_3353 (FGE) as an obligatory dependency of any BetC-containing module.
- **S1 uptake → CANDIDATE_UNCERTAIN:** retain PP_0075 and PP_0076 as candidates but flag substrate uncertainty; do not assert a specific COS transporter.
- **Regulation:** add PP_0079 as candidate regulator (module_needs_revision if a regulatory step is expected but absent from the generic module).

- **Module boundary:** keep COS-utilization (import + desulfation) **distinct** from the downstream choline→betaine oxidation module and from general sulfur-ester scavenging; the generic module should **not** absorb *betA/betB* or the Cys pathway.
- **GO requests:** existing GO terms (GO:0047753 choline-sulfatase; GO:0015871 choline transport) suffice; a dedicated "choline-O-sulfate transmembrane transporter activity" child term would help disambiguate PP_0075/PP_0076 if such a term is lacking.

7. Genes to promote to full review

1. **PP_0077 (*betC*)** — anchor desulfation gene; verify motif, FGE dependency, and *S. meliloti* orthology. (*highest priority*)
2. **PP_0075** — correct/qualify the "choline sulfate transporter" annotation; determine SulP substrate.
3. **PP_0076** — confirm as COS/choline SBP and identify its cognate ABC permease/ATPase.
4. **PP_0079** — assess as pathway-specific transcriptional regulator.
5. **PP_3353 (FGE)** — record as shared maturation dependency.

8. Key references

- Osterås M, Boncompagni E, Vincent N, Poggi M-C, Le Rudulier D. *Presence of a gene encoding choline sulfatase in Sinorhizobium meliloti bet operon: choline-O-sulfate is metabolized into glycine betaine*. PNAS 1998.

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— Defines BetC/choline-sulfatase (EC 3.1.6.6); COS used as sole C/N/S source via a functional *bet* locus. (*Strong transfer to KT2440: same enzyme family/motif.*)

- Nau-Wagner G, Boch J, Le Good JA, Bremer E. *High-affinity transport of choline-O-sulfate and its use as a compatible solute in Bacillus subtilis*. Appl Environ Microbiol 1999.

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— COS acquired via the OpuC ABC transporter; informs the uptake-mechanism question. (*Mechanistic, transporter-family divergent from PP_0075.*)

- Holt S, Kankipati H, De Graeve S, et al. *Major sulfonate transporter Soa1 in Saccharomyces cerevisiae and considerable substrate diversity in its fungal family*. 2017.

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— Establishes COS/sulfonates as prominent environmental sulfur sources (ecological rationale).

- UniProt/InterPro/KEGG (taxon 160488): Q88RQ2 (PP_0077 betC), Q88RQ3 (PP_0076), Q88RQ4 (PP_0075), Q88HK3 (PP_3353 FGE); InterPro IPR017785 (Choline-sulfatase), IPR001902 (SLC26A/SulP). Accessed 2026-08-31. (*Computational evidence.*)

Uncertainty statement: desulfation (BetC) assignment is **high-confidence homology + sequence motif**; uptake assignments are **low-to-medium confidence, computational only**; no direct KT2440 experimental data exist. Resolving experiments: growth of KT2440 and $\Delta betC/\Delta PP_0075/\Delta PP_0076$ mutants on COS as sole sulfur (and C/N) source; ^{14}C -COS transport assays; and induction/RNA-seq of PP_0075–0079 under sulfate limitation or COS exposure.